ULMIS — Complete System Specification

Field	Value
Product	ULMIS — University Library Management Information System
Document type	Complete System Specification (SRS + Architecture + Features + LLD)
Date	August 2026
Classification	Proprietary

Document contents

1. System Requirements Specification (SRS)
2. Architecture Design
3. Features & Requirements
4. Low-Level Design (LLD) Workflows

Part 1 — System Requirements Specification (SRS)

University Library Management Information System (ULMIS)

Field	Value
Document ID	SRS-ULMIS-001
Product	University Library Management Information System (ULMIS / LibraStack)
Repository	`LibraryManagementSystem`
Version	1.0
Date	August 2026
Classification	Proprietary

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1. Introduction

1.1 Purpose

This SRS defines requirements for **ULMIS**, a single-institution university library platform for cataloging, inventory, members, circulation, OPAC/self-service, digital holdings, kiosk platform, queue/locator boards, reporting, notifications, and campus integration adapters.

1.2 Scope

In scope (software)

- Modules 2.1–2.9 and 2.11–2.15 per ULMIS requirements matrix
- Attendance logging and staff Users/Settings
- Staff web, public OPAC, and kiosk clients on one REST API
- Borrow blocks and six mandated report exports
- Local JWT auth and University Platform Access SSO (`AUTH_MODE` local | dual | platform)

Explicitly excluded

- **§2.10 RFID Library Automation** (gates, handhelds, RFID checkout) — hardware/vendor out of this delivery
- Biometric/smartcard wall appliances, turnstile hardware, 55" display hardware
- Live LDAP binds / unverified SSO assertions (return 501 until verified)
- Certified ≥300 concurrent / ≤3s proof (architecture ready; load test separate)

1.3 Definitions

Term	Definition
OPAC	Online Public Access Catalog
Copy	Physical inventory unit (barcode-level circulation)
Kiosk Platform	Staff admin + self-service terminal software
University mode	Single-institution deployment (default)

1.4 References

- `LibraryManagementSystem/README.md`
- `docs/compliance/ULMIS_SCOPE.md`
- `docs/compliance/ULMIS_REQUIREMENTS_MATRIX.md`
- `docs/compliance/ULMIS_NFR.md`
- `docs/ULMIS-Complete-Guide.md`

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2. Overall Description

2.1 Product Perspective

App	Port (dev)	Audience	Auth
Staff web (`apps/web`)	5173	Librarians, admins	JWT
Public OPAC (`apps/opac`)	5174	Public / patrons	Tenant-scoped read
Kiosk (`apps/kiosk`)	5175	Self-service terminals	Device code + API key
API (`apps/api`)	3333	All clients	JWT / kiosk / public

Default: `DEPLOYMENT_MODE=university` , `ALLOW_PUBLIC_REGISTER=false` , fixed `DEFAULT_TENANT_SLUG` .

2.2 User Classes

Class	Capabilities
Administrator	Full tenant permissions
Librarian	Catalog, members, circulation, fines, digital, reports
Staff	Desk checkout/return/reservations
Member (portal)	Loans, renew, reservations, reading lists, digital access
Public OPAC user	Read-only search/citations; queue board without PII

Kiosk patron	Borrow/return/account/history via device credentials

Member types: `student` | `faculty` | `employee` | `researcher` | `guest` | `alumni`

2.3 Constraints

- Not multi-customer SaaS by default
- RFID excluded from readiness claim
- Public OPAC read-only; public queue must not expose patron identity
- Kiosk API keys hashed at rest; plaintext shown once on create/rotate
- Fines in Philippine pesos (₱)

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3. Functional Requirements

3.1 Core Modules

Module	ID range	Key requirements	Status
Dashboard & monitoring	FR-2.1.*	KPIs, widgets, summary PDF	Ready
Catalog	FR-2.2.*	MARC/Dublin Core fields; CSV/MARC JSON/Excel import; versions	Ready
Item & inventory	FR-2.3.*	Copies, barcodes, statuses, locate, history	Ready
User management	FR-2.4.*	Members, staff users, RBAC, digital card; ARIS/UHRIS adapters	Ready
Circulation	FR-2.5.*	Desk + kiosk borrow/return/renew; fines; reservations	Ready
OPAC	FR-2.6.*	Staff + public OPAC; citations; reading lists; page size cap 25	Ready
Digital repository	FR-2.7.*	Create/list/search; view/download; PDF view	Ready
Digital resources	FR-2.8.*	E-resource registry + discovery	Ready
E-library portal	FR-2.9.*	`/portal` loans/fines/lists; `/digital`	Ready
RFID	FR-2.10.*	**Excluded**	Excluded
Self-service kiosk	FR-2.11.*	Admin platform + kiosk client; device keys; idle timeout	Ready
Resource locator	FR-2.12.*	Shelf locations; locate UI	Ready

Queue management	FR-2.13.*	Issue/call; public board without PII	Ready
Reporting	FR-2.14.*	Six mandated reports CSV/PDF	Ready
Notifications	FR-2.15.*	In-app; SMTP/Twilio when configured	Ready
Attendance	FR-ATT.*	Card visit log + daily summary	Ready

3.2 Borrow Blocks (Must Enforce)

ID	Block condition
FR-BLK-001	Not enrolled / inactive academic status
FR-BLK-002	Blacklisted
FR-BLK-003	Overdue items
FR-BLK-004	Unpaid fines/charges
FR-BLK-005	Suspended or expired membership

3.3 Mandated Reports

Report	Endpoint
Collections	`GET /api/v1/reports/collections`
Borrowed books	`GET /api/v1/reports/borrowed`
Overdue books	`GET /api/v1/reports/overdue`
Book ledger	`GET /api/v1/reports/book_ledger`
Patron ledger	`GET /api/v1/reports/patron_ledger`
Charges / penalties	`GET /api/v1/reports/charges`

Exports require `reports:export`. Scheduled email when SMTP and queue worker are enabled.

3.4 Security Rules (Functional)

ID	Requirement	Priority
FR-SEC-001	No public write for circulation/member data without kiosk device credentials	Must
FR-SEC-002	Public OPAC tenant-scoped and read-only	Must
FR-SEC-003	Integration sync admin-only, batch-limited, card-ID match	Must

FR-SEC-004	Do not accept unverified directory passwords or SSO assertions in stubs	Must

4. External Interface Requirements

Interface	Detail
REST API	`/api/v1/*` ; Swagger `/api/v1/docs`
University Platform Access SSO	`PLATFORM_*` env; platform codes for Library stack
ARIS / UHRIS	Inbound patron sync adapters (credentials required for live pull)
SMTP / Twilio	Notifications when configured
Kiosk APIs	`/api/v1/hardware/kiosks*` , `/api/v1/kiosk/*`

5. Non-Functional Requirements

ID	Requirement	Notes
NFR-G4-001	≥300 concurrent users; ≤3s standard transactions	Target; not load-proven in-repo
NFR-G4-002	Stateless JWT API; Postgres + Redis	Ready
NFR-OPS-001	CI typecheck/build; Redis for rate limits and jobs	Documented
NFR-OPS-002	Managed Postgres backups; `DB_SSL=true` for hosted	Ops checklist
NFR-SEC-001	CORS includes staff, OPAC, and kiosk origins	Must

6. Verification Criteria (Summary)

Scenario	Expected result
University bootstrap	Admin login without org slug field
Checkout with overdue/fines	Blocked per FR-BLK rules
Public OPAC	Read-only search; no member writes
Kiosk borrow/return	Works with device credentials
Mandated reports	CSV/PDF export with permission
RFID claim	Not asserted for this delivery

7. Document Control

Version	Date	Description
1.0	August 2026	SRS from ULMIS compliance matrix and guides

Related design docs

- **ULMIS-Architecture.md** — Architecture design (multi-client, domains, security)
- **ULMIS-Features-Requirements.md** — Feature catalog + requirements matrix
- **ULMIS-LLD.md** — Low-level design workflows (circulation, borrow blocks, kiosk, OPAC/queue, reports)

End of SRS — ULMIS

Part 2 — Architecture Design

ULMIS — University Library Management Information System

Field	Value
Document ID	AD-ULMIS-001
Product	ULMIS
Companions	LibraryManagementSystem-SRS.md · ULMIS-Features-Requirements.md · ULMIS-LLD.md
Date	August 2026

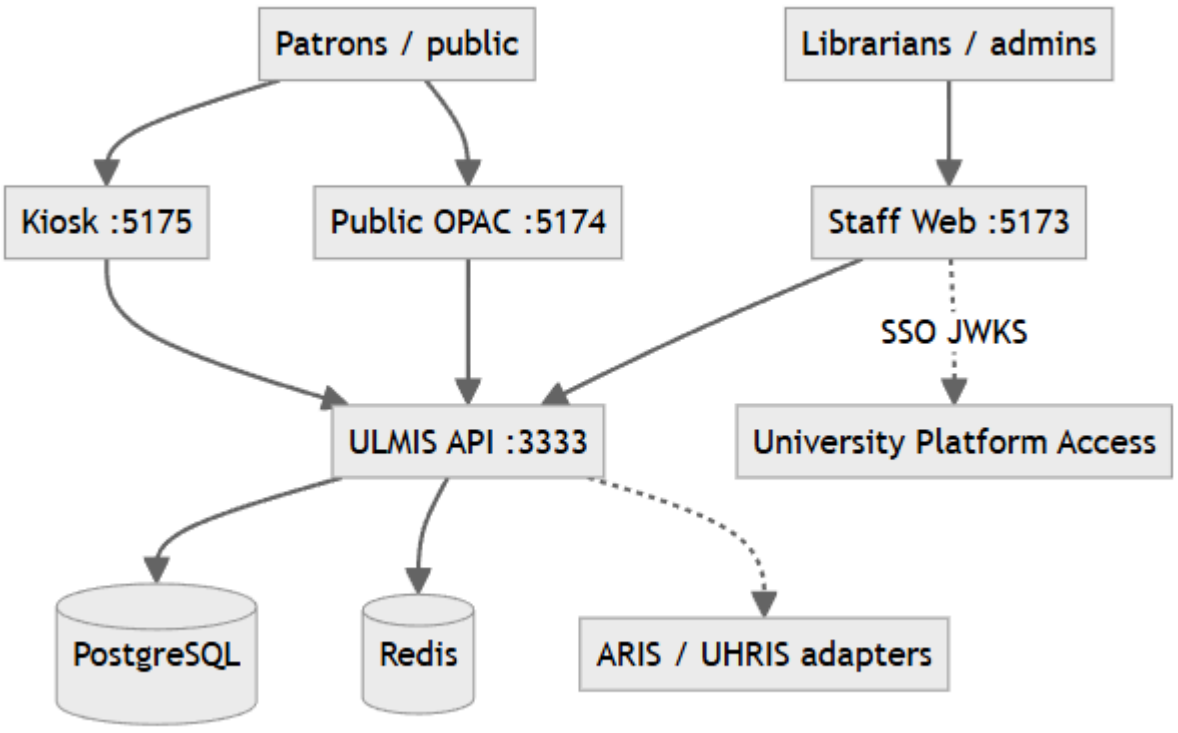
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1. Purpose

Architecture Design for single-institution university library mode: one API, three client apps (staff, OPAC, kiosk).

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2. System context



3. Container view

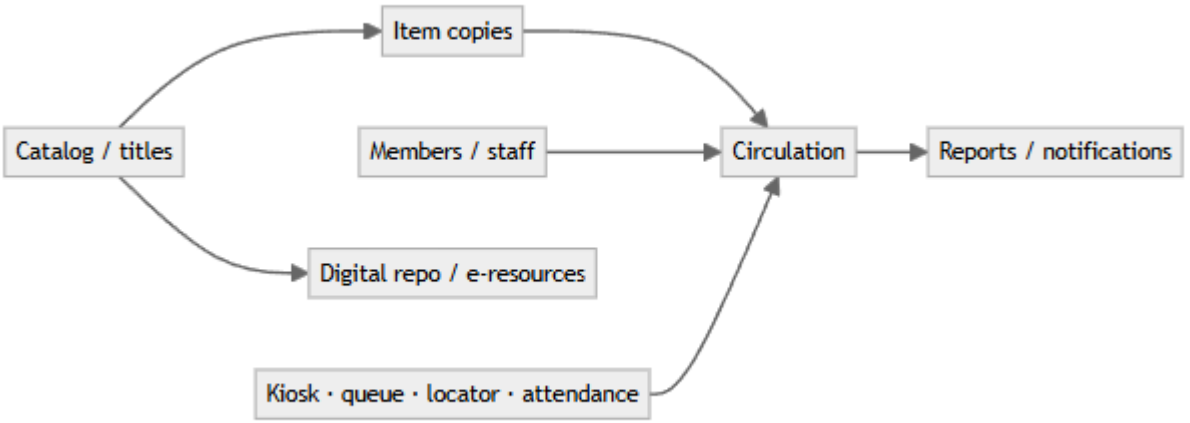
Container	Auth model	Capabilities
Staff web	JWT + RBAC	Catalog, circulation, members, reports, kiosk admin
Public OPAC	Tenant-scoped read	Search, citations, queue board (no PII)
Kiosk	Device code + API key	Borrow/return/account/history
API	JWT / kiosk / public	`/api/v1/*`
Postgres	—	Bibliographic, copies, loans, members
Redis	—	Rate limits, scheduled report jobs

Deployment mode: `DEPLOYMENT_MODE=university`, `ALLOW_PUBLIC_REGISTER=false`, fixed `DEFAULT_TENANT_SLUG`.

4. Shared packages

Package	Role
`@librastack/client-config`	Shared Vite client helpers
`@librastack/permissions`	Shared RBAC constants

5. Domain architecture



Domain	Notes
Catalog	MARC/Dublin Core fields; bulk import
Copies	Barcode-level status & history
Circulation	Desk + kiosk; fines ₱; reservations; borrow blocks
Digital	Repository + e-resource registry
Hardware software	Kiosk platform APIs; RFID **excluded**

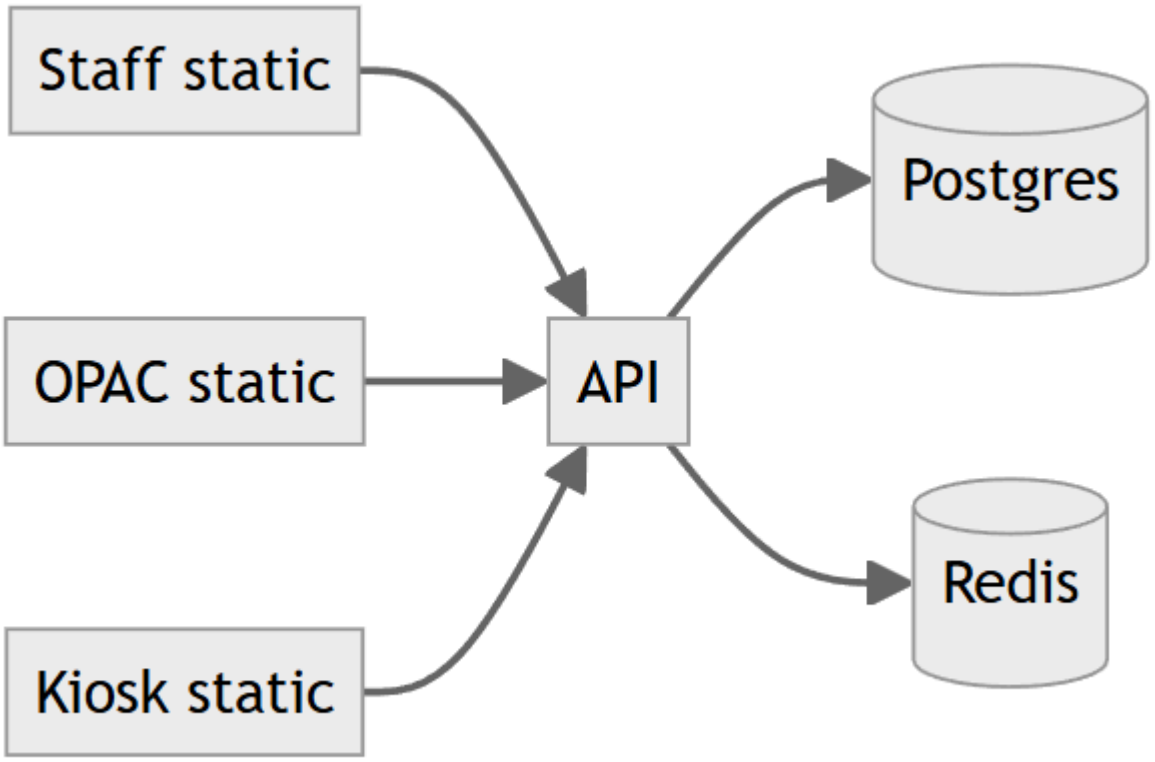
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6. Security architecture

Rule	Enforcement
No public write for circulation/members	Kiosk writes require device credentials
Public OPAC read-only	Tenant-scoped
Queue board	Strip patron identity
Report export	`reports:export`
Sync	Admin-only, batch-limited
Kiosk keys	Hashed at rest; plaintext once

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7. Deployment topology



Supports on-prem, hybrid, or cloud. CORS must include staff, OPAC, and kiosk origins.

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8. Integration architecture

Peer	Pattern
University Platform Access	`AUTH_MODE` local|dual|platform; JWKS
ARIS / UHRIS	Inbound patron sync adapters
SMTP / Twilio	Notifications when configured
LDAP/AD	Stubbed until verified (501)

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9. Quality attributes

Attribute	Approach
Concurrency target	≥300 users / ≤3s (prove in staging)
HA readiness	Stateless JWT API + managed Postgres/Redis

University scope	Single tenant; not multi-customer SaaS by default

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End of Architecture Design — ULMIS

Part 3 — Features & Requirements

ULMIS — University Library Management Information System

Field	Value
Document ID	FR-ULMIS-001
Product	ULMIS
Companions	LibraryManagementSystem-SRS.md · ULMIS-LLD.md
Date	August 2026

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1. Feature summary

Module	Features	Status
2.1 Dashboard	KPIs, widgets, summary PDF	Ready
2.2 Catalog	MARC/Dublin Core fields; CSV/MARC JSON/Excel import; versions	Ready
2.3 Inventory	Copies, barcodes, statuses, locate, history	Ready
2.4 Users	Members, staff RBAC, digital card; ARIS/UHRIS adapters	Ready
2.5 Circulation	Desk + kiosk borrow/return/renew; fines; reservations	Ready
2.6 OPAC	Staff + public OPAC; citations; reading lists	Ready
2.7 Digital repository	Create/list/search; view/download; PDF view	Ready
2.8 Digital resources	E-resource registry + discovery	Ready
2.9 E-library portal	Loans, fines, lists, digital access	Ready
2.10 RFID	—	**Excluded**
2.11 Kiosk	Admin platform + self-service client	Ready
2.12 Locator	Shelf locations; locate UI	Ready
2.13 Queue	Issue/call; public board without PII	Ready
2.14 Reporting	Six mandated reports CSV/PDF	Ready
2.15 Notifications	In-app; SMTP/Twilio when configured	Ready
Attendance	Card visit log + daily summary	Ready

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2. Feature catalog

Feature ID	Feature	Description	Priority
F-2.1-01	Dashboard KPIs	Loans, overdue, requests, attendance	Must
F-2.1-02	Widget toggles + PDF summary	Configurable widgets; export	Should
F-2.2-01	Bibliographic catalog	MARC/Dublin Core–compatible fields	Must
F-2.2-02	Bulk import	CSV, MARC JSON, Excel	Must
F-2.2-03	Catalog versions	Version history panel	Should
F-2.2-04	Authority control	Authors, subjects, publishers authority lists; batch update/delete with active-loan safeguards	Should
F-2.3-01	Copy inventory	Barcodes, statuses, locate, history	Must
F-2.3-02	Inventory audit	Barcode scan audits + discrepancy reports (RFID hardware excluded)	Should
F-2.4-01	Member management	Register, approve, digital card	Must
F-2.4-02	Staff users + RBAC	Invite/edit roles	Must
F-2.4-03	Campus sync adapters	ARIS/UHRIS inbound mapping	Should
F-2.5-01	Desk circulation	Checkout, return, renew, mark lost	Must
F-2.5-02	Fines (₱)	Assess and manage charges	Must
F-2.5-03	Reservations	Place/fulfill holds	Must
F-2.5-04	Borrow blocks	Enforce eligibility rules	Must
F-2.5-05	Due dates & penalties	Auto compute due dates and penalties from policy	Must
F-2.6-01	Public OPAC	Read-only search (page size ≤25)	Must
F-2.6-02	Citations & reading lists	Generate citations; lists	Should
F-2.7-01	Digital repository	Access-controlled digital items	Must
F-2.8-01	E-resources	Registry + discovery UI	Must
F-2.9-01	Member portal	Loans, fines, lists, digital	Must
F-2.11-01	Kiosk Platform admin	Devices, keys, enable/disable, logs	Must
F-2.11-02	Self-service kiosk	Borrow/return/account/history	Must

F-2.11-03	Kiosk session timeout	Self-service idle timeout	Must
F-2.11-04	Server time only	Use API/server clock; never trust kiosk local clock	Must
F-2.12-01	Resource locator	Shelf locations + locate UI (software; 55" board hardware excluded)	Should
F-2.13-01	Queue management	Staff console + public board	Must
F-2.14-01	Mandated reports	Six reports; CSV/PDF; schedule email	Must
F-2.14-02	Print layout	PDF export with user-defined page size/orientation where printer allows	Should
F-2.15-01	Notifications	In-app + optional SMTP/SMS	Must
F-ATT-01	Attendance log	Card visits + dashboard count	Should
F-AUTH-01	Local JWT + UPA SSO	University login / dual SSO	Must
F-AUTH-02	LDAP/AD campus SSO	Deferred until campus directory verified (HTTP 501); UPA OAuth SSO is primary	Could

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3. Requirements matrix

Req ID	Requirement	Related features	Priority
FR-2.1.*	Dashboard monitoring	F-2.1-*	Must
FR-2.2.*	Catalog + import	F-2.2-*	Must
FR-2.3.*	Copy-level inventory	F-2.3-01	Must
FR-2.4.*	Members + staff RBAC	F-2.4-*	Must
FR-2.5.*	Circulation + fines + reservations	F-2.5-*	Must
FR-BLK-001..005	Borrow blocks (enrolled, blacklist, overdue, fines, membership)	F-2.5-04	Must
FR-2.6.*	OPAC read-only public + staff	F-2.6-*	Must
FR-2.7.* / FR-2.8.*	Digital repo + e-resources	F-2.7-01, F-2.8-01	Must
FR-2.9.*	E-library portal	F-2.9-01	Must
FR-2.10.*	RFID	—	Excluded

FR-2.11.*	Kiosk platform + client	F-2.11-*	Must
FR-2.12.*	Locator	F-2.12-01	Should
FR-2.13.*	Queue without PII on public board	F-2.13-01	Must
FR-2.14.*	Six mandated reports + `reports:export`	F-2.14-01	Must
FR-2.15.*	Notifications	F-2.15-01	Must
FR-ATT.*	Attendance	F-ATT-01	Should
FR-SEC-001..004	Public write ban; OPAC read-only; sync admin-only; no unverified SSO	—	Must
FR-2.2-AUTH	Authority control + batch safeguards	F-2.2-04	Should
FR-2.5-DUE	Auto due dates and penalties	F-2.5-05	Must
FR-2.11-TIME	Kiosk timeout + server clock	F-2.11-03, F-2.11-04	Must
FR-AUTH-LDAP	LDAP/AD until verified	F-AUTH-02	Could

Source DOCX traceability (Library-Management-Information-System.docx) — 100%

Source requirement	Feature / ID	Status
Web LMS modular modules 2.1–2.9, 2.11–2.15	F-2.* summary	Delivered
≥300 concurrent / ≤3s	NFR-G4-001	Delivered (target)
MARC21 / Dublin Core / bulk CSV-Excel-MARC	F-2.2-01, F-2.2-02	Delivered
Authority control + version history	F-2.2-03, F-2.2-04	Delivered
Item/copy inventory + statuses + history	F-2.3-01	Delivered
Borrow blocks (unenrolled, blacklist, overdue, unpaid)	F-2.5-04	Delivered
Auto due dates & penalties	F-2.5-05	Delivered
Six mandated reports	F-2.14-01	Delivered
OPAC + portal + digital repo/resources	F-2.6..2.9	Delivered
Kiosk self-service + timeout + server time	F-2.11-*	Delivered
Queue management (no public PII)	F-2.13-01	Delivered
Notifications email/SMS/in-app	F-2.15-01	Delivered
ARIS / UHRIS sync adapters	F-2.4-03	Delivered
Entry/attendance monitoring	F-ATT-01	Delivered
SSO via University Platform Access	F-AUTH-01	Delivered

LDAP/AD campus SSO	F-AUTH-02	Excluded (until verified)
RFID module / gates / handhelds	FR-2.10.*	Excluded
Biometric/smartcard wall kiosk hardware	—	Excluded
55" interactive locator board hardware	F-2.12-01 note	Excluded
Nationwide SaaS marketplace mode	University single-tenant	Excluded
PDF user-defined page size/orientation	F-2.14-02	Delivered
Training / install / UAT services	SVC-ULMIS-*	Service

Mandated reports

Report	Endpoint
Collections	`GET /api/v1/reports/collections`
Borrowed	`GET /api/v1/reports/borrowed`
Overdue	`GET /api/v1/reports/overdue`
Book ledger	`GET /api/v1/reports/book_ledger`
Patron ledger	`GET /api/v1/reports/patron_ledger`
Charges	`GET /api/v1/reports/charges`

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4. Non-functional requirements (summary)

ID	Requirement	Notes
NFR-G4-001	≥300 concurrent; ≤3s transactions	Target; load-prove in staging
NFR-G4-002	Stateless JWT; Postgres + Redis	Ready
NFR-OPS-001	CI; Redis for limits/jobs	Documented
NFR-SEC-001	CORS for staff, OPAC, kiosk	Must

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5. Actors & feature access

Actor	Primary features
Administrator	All ready modules
Librarian	Catalog, members, circulation, digital, reports

Desk staff	F-2.5-, *F-2.13*-
Member	F-2.9-01, portal renew/holds
Public	F-2.6-01, queue board
Kiosk patron	F-2.11-02

End of Features & Requirements — ULMIS

Part 4 — Low-Level Design (LLD) Workflows

ULMIS (University Library Management Information System)

Field	Value
Document ID	LLD-ULMIS-001
Product	ULMIS
Companion	LibraryManagementSystem-SRS.md
Date	August 2026

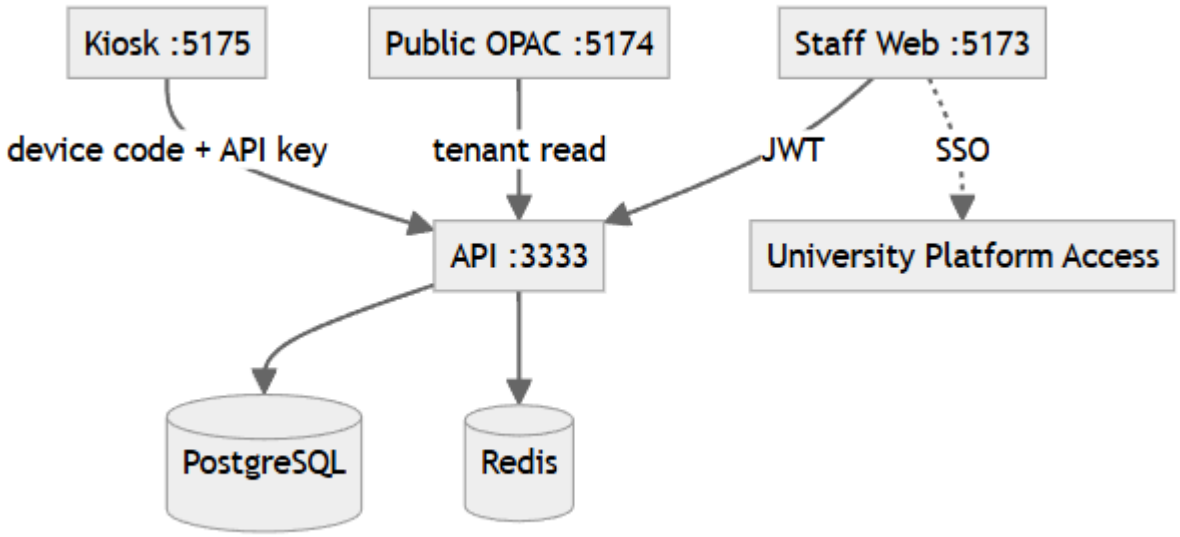
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1. Purpose

This LLD documents circulation, OPAC, kiosk, queue, borrow-block enforcement, and reporting workflows for university-mode ULMIS.

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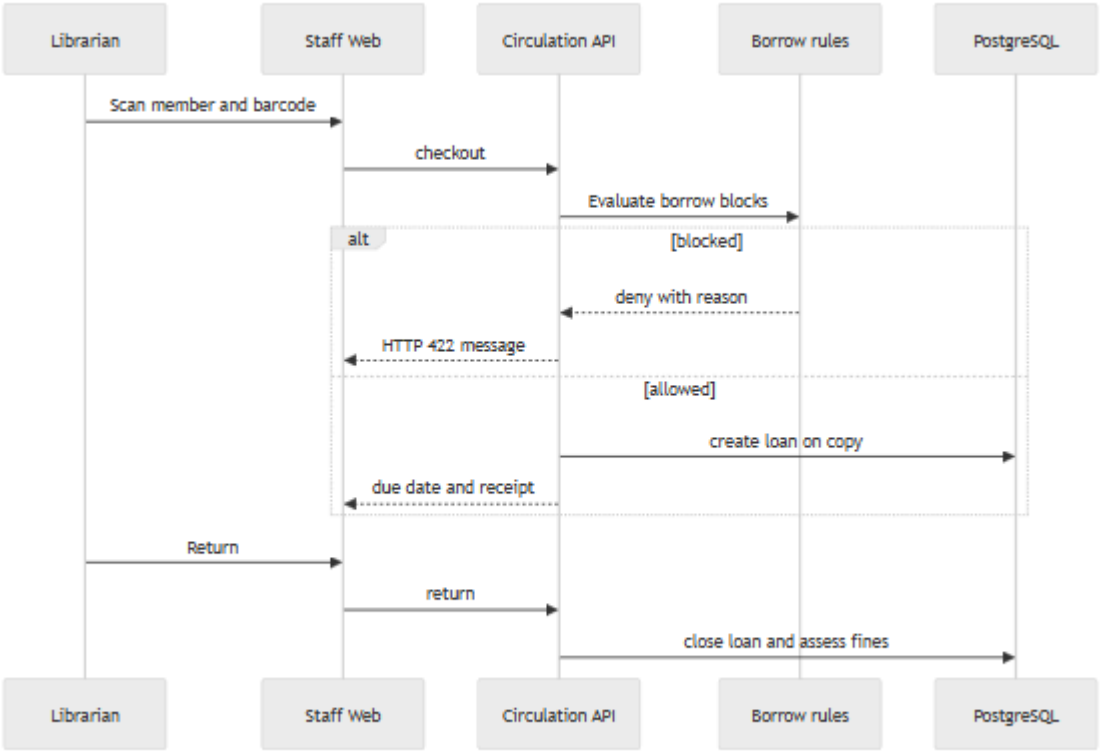
2. System context (multi-client)



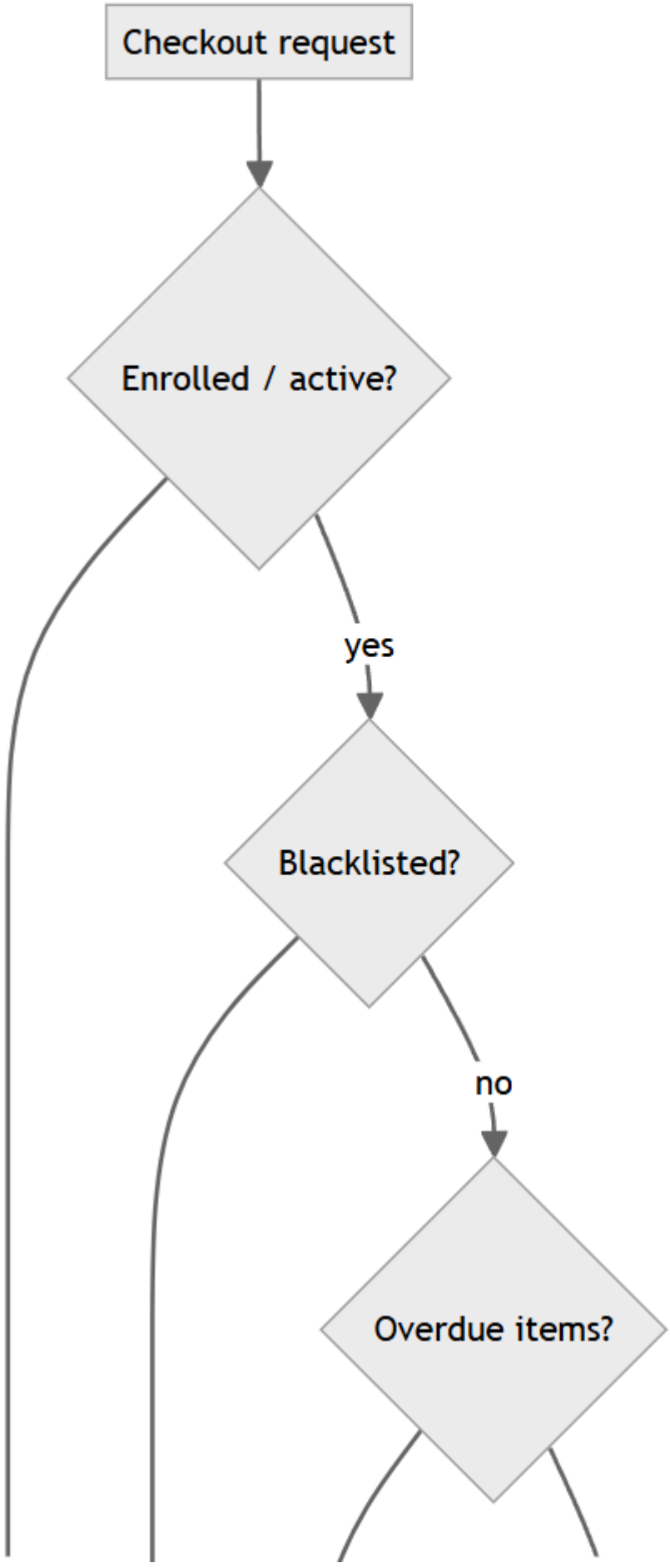
Client	Write circulation?	Auth
Staff web	Yes	JWT + RBAC

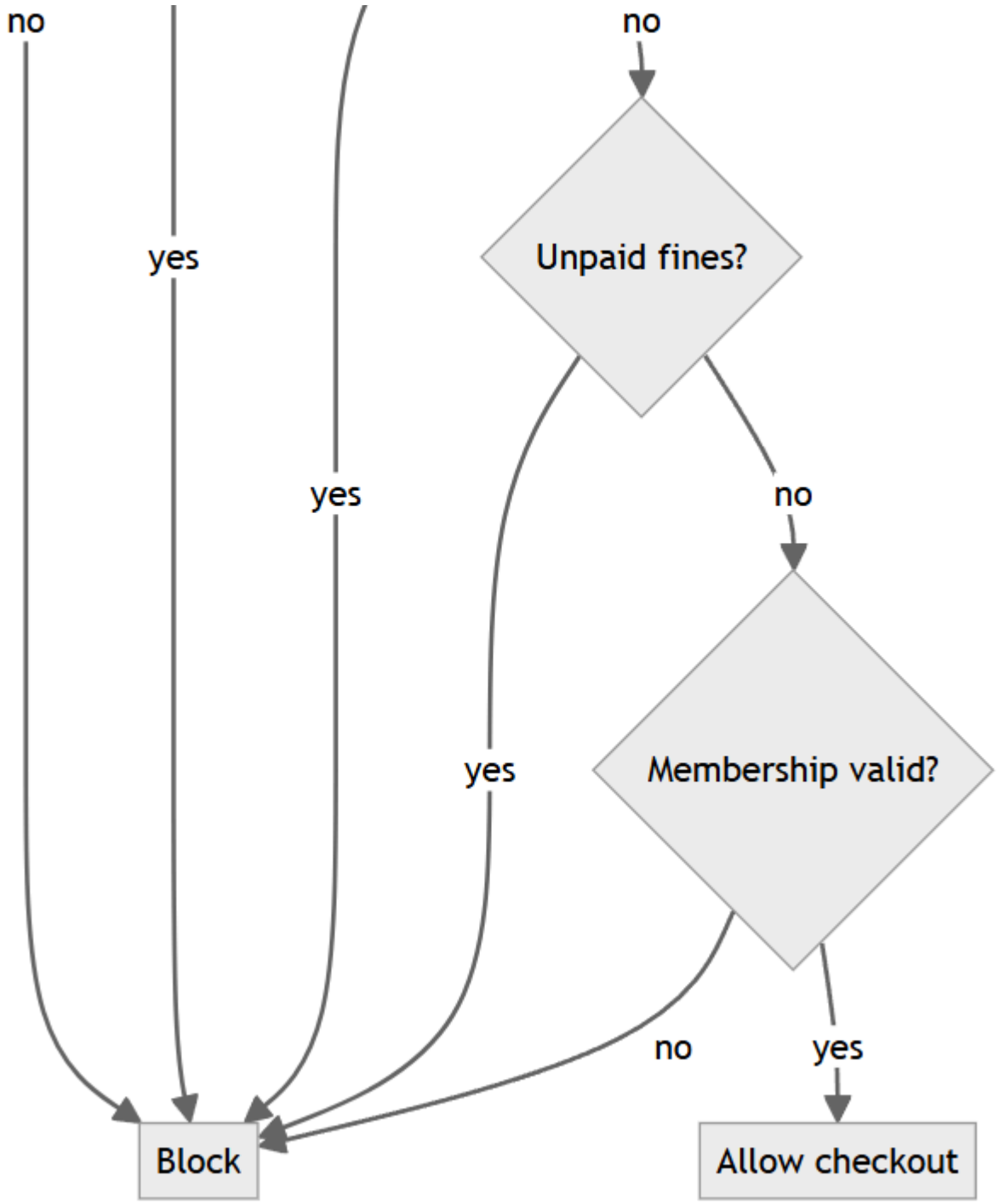
Public OPAC	No	Tenant-scoped read
Kiosk	Yes (self-service)	Device credentials

3. Staff circulation workflow

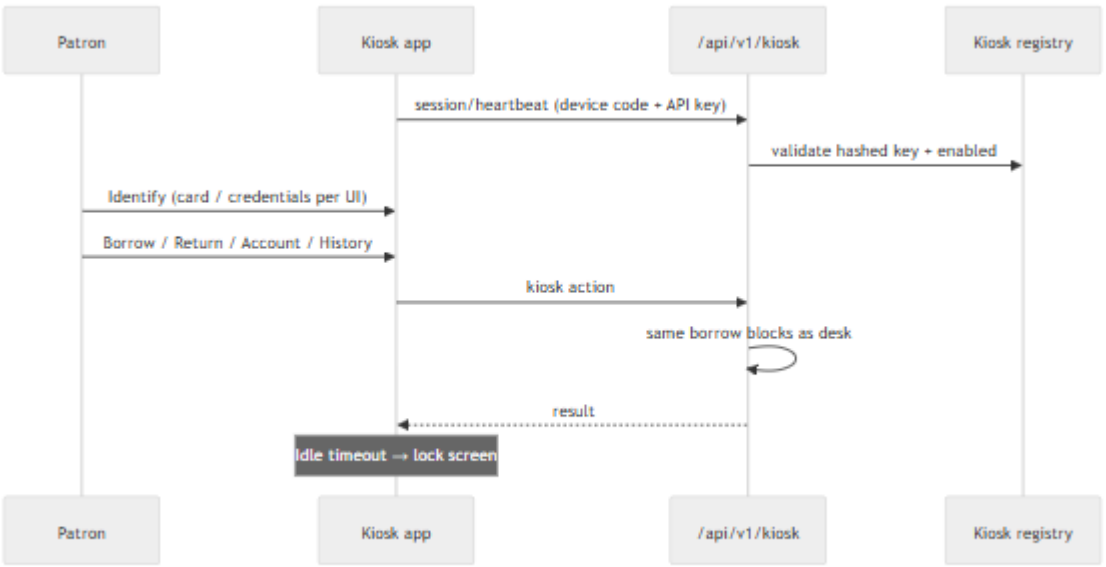


4. Borrow blocks (decision tree)


Checkout request
Enrolled / active?
yes
Blacklisted?
no
Overdue items?



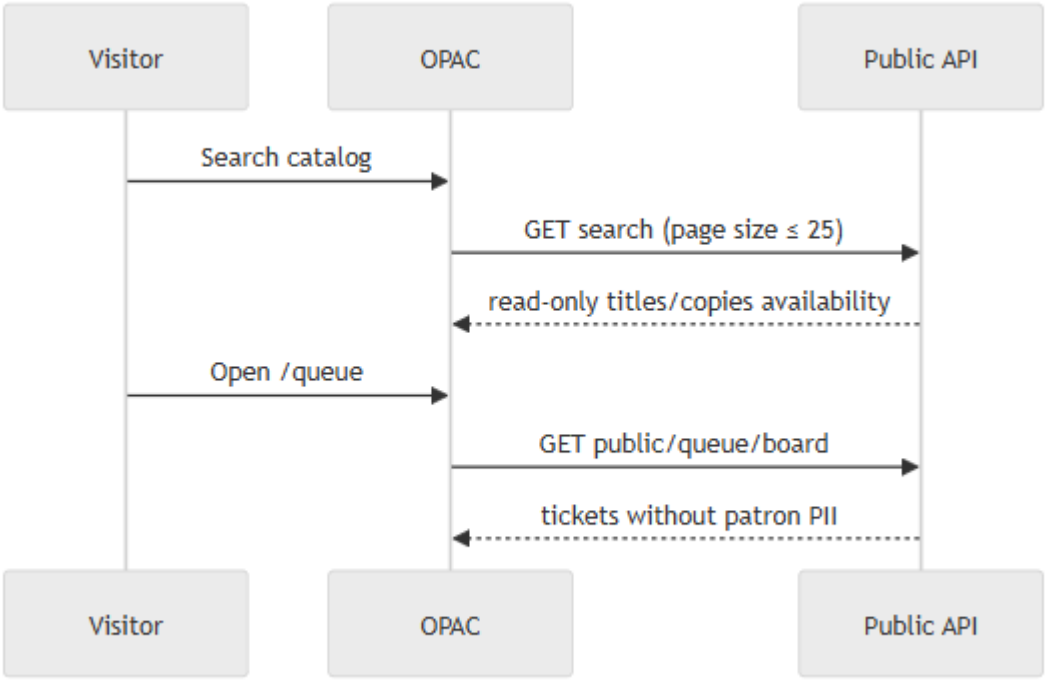
5. Kiosk self-service workflow


Patron
Kiosk app
/api/v1/kiosk
Kiosk registry
session/heartbeat (device code + API key)
validate hashed key + enabled
Identify (card / credentials per UI)
Borrow / Return / Account / History
kiosk action
same borrow blocks as desk
result
Idle timeout → lock screen
Patron
Kiosk app
/api/v1/kiosk
Kiosk registry

Admin (`/kiosk-platform`): register device, enable/disable, rotate key (plaintext once), view logs.

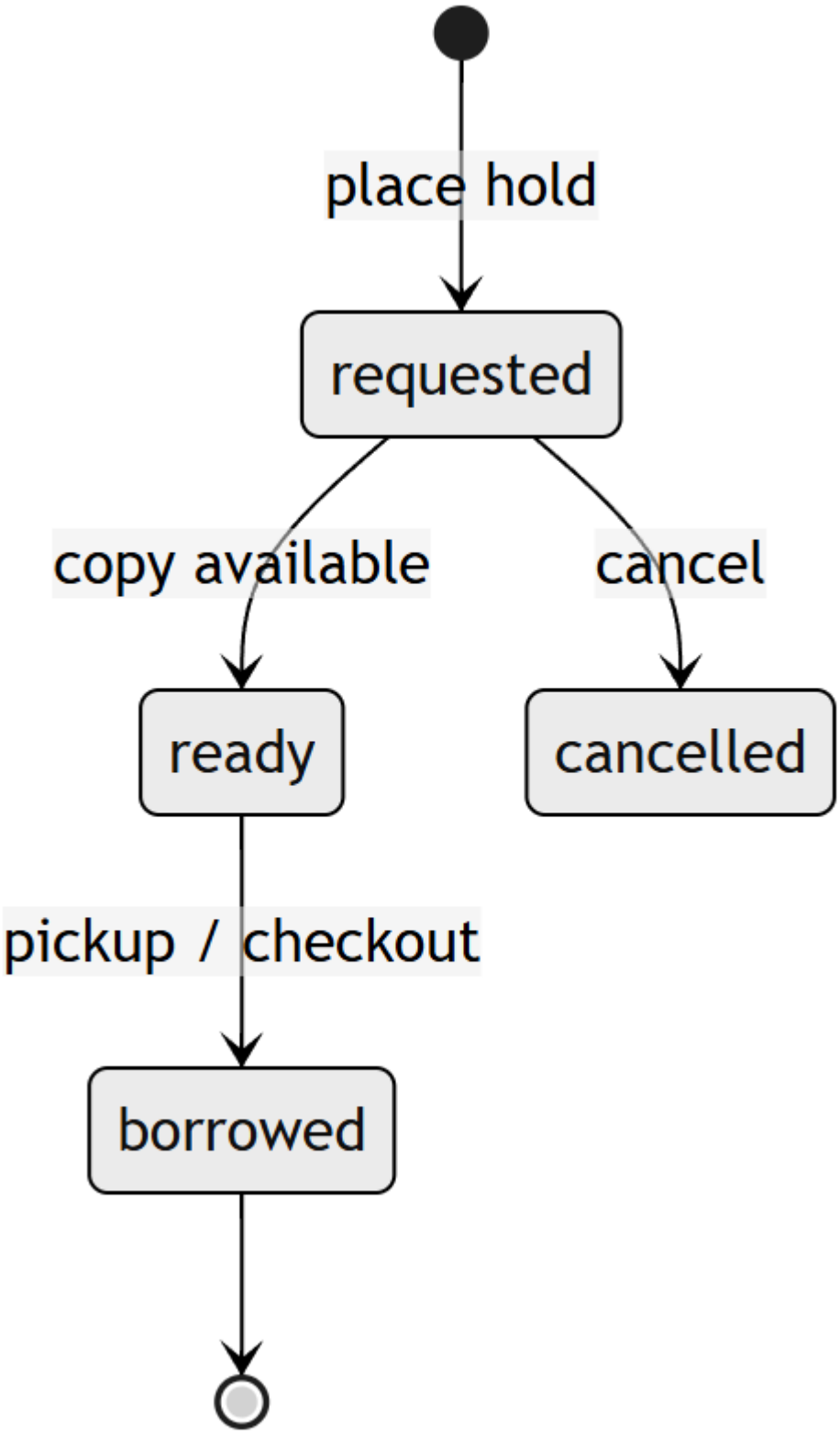
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6. Public OPAC & queue board



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7. Reservation & renew



Renew: staff or portal/kiosk renew if policy allows and no blocks/holds conflict.

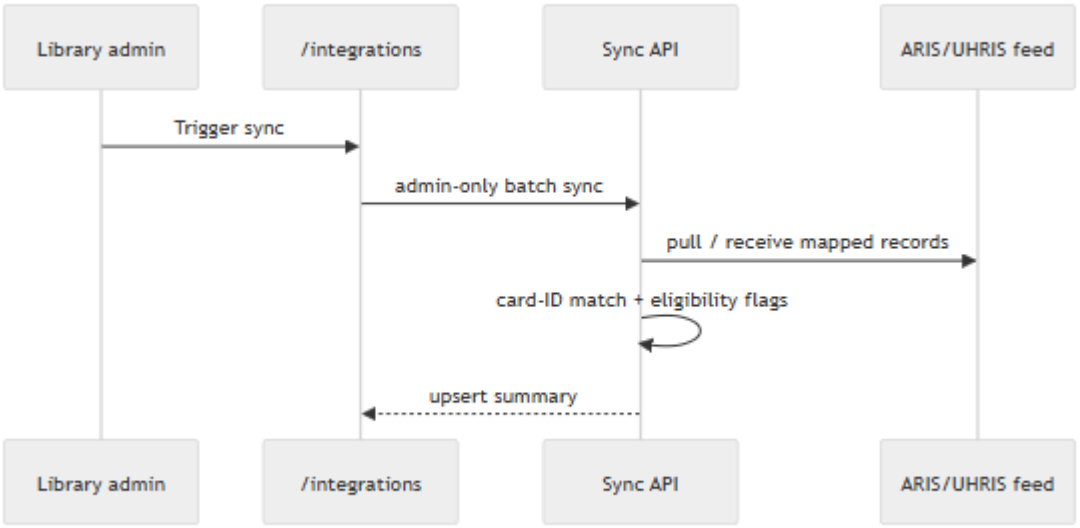
8. Mandated report generation



Reports: collections · borrowed · overdue · book_ledger · patron_ledger · charges.

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9. Member sync (ARIS / UHRIS adapters)



LDAP/SSO campus stubs return **501** until verified.

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10. Error / edge cases

Case	Expected behavior
Public write to circulation	Denied
Invalid kiosk key	401; no plaintext key recovery
Queue board	No patron identity exposed
RFID paths	Out of scope for this delivery
CORS missing OPAC/kiosk origin	Browser blocked

End of LLD — ULMIS